A Logarithmic Market Scoring Rule Agent-based Model to Evaluate Prediction Markets*

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Abstract

Prediction Markets (PMs) are markets in which agents trade event contingent assets. Enterprises use PMs to forecast revenues and project deadlines. This paper presents an Agent-based model, called Logarithmic Market Scoring Rule-Automated Market Maker (LMSR-ASM), to evaluate Prediction Markets. Our model is capable of testing different types of Automated Market Makers (AMMs), which are mathematical functions or computational mechanisms needed to provide liquidity in Prediction Markets. The model offers insights into how to set parameters in a PM and how profits react to contrasting settings and AMMs. In addition, we simulate different probability processes, distinct AMMs, and agent behaviors. This paper also utilizes the LMSR-ASM to evaluate the impact of choosing initial prices in profits and revenue opportunities regarding AMM computational implementation. We show that we can use the LMSR-ASM to find optimal parameters for maximizing profits in PMs and how different AMMs affect market results under a variety of settings.

Keywords: Agent Based Model, Prediction Market.
JEL Code: D8, G1 .

*The source code is at `https://github.com/Athosvcc/LMSR-ASM`.

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1. Introduction

Prediction Markets (PMs), also called Information Markets, are forums for trading contracts that yield payments based on the outcome of uncertain events (Wolfers & Zitzewitz, 2016). A more intuitive explanation for a PM is a market for assets that pay depending on the realization of a given event. Situations like "Will it rain next month?" or "Who will be elected president in X country?" illustrates possible PMs. For the former, the assets are "Yes" and "No". For the latter, it is one for each candidate. A binary PM is a market for an event with two possible options. One may create a PM for any event with finite outcomes.

Corporate environments extensively use PMs. Circumstances that allow the usage of PMs are: to forecast the number of sales of a determined product, future revenue, and whether projects will accomplish the proposed timeline with great success (see Arrow et al. (2008) for examples, and Cowgill & Zitzewitz (2015) for an evaluation). There are also sites available to the general public, such as *PredictIT*[1], the *Iowa Electronic Markets*[2], and *Betfair*.[3] These platforms contain multiple markets, which are mostly related to politics. Moreover, we mention the DeFI[4] apps. It allows users to create their PMs using the *Augur*[5] and *Gnosis*[6] protocols, working on the Ethereum blockchain.

One difficulty inherent to PMs is their few numbers of participants. This fact makes liquidity a hindrance to trade and, therefore, aggregate forecasts. Continuous double auctions and order books, for instance, would only be efficient in a market with a volume of trades rarely observed in PMs. The first is a system that executes trades by matching the compatible sell and buy orders in terms of price (Das et al., 2001). The second is a collection of price contingent buy and sell orders (Roşu, 2009), as used by traditional stock exchanges (Pennock, 2004). Likewise, the use of human market makers would raise costs for the operation of PMs. Thus, PMs need a cost-effective way to create liquidity for their assets. This necessity leads to Automated Market Makers (AMMs), which are mathematical functions or computational mechanisms capable of providing liquidity automatically, and handling buy and sell orders in a

[1] `https://predicit.org`.

[2] `https://iemweb.biz.uiowa.edu/`.

[3] `https://betfair.com`.

[4] A Decentralized Finance (DeFi) Application is defined by Wang (2020) as a smart contract stored in a public distributed ledger (such as a blockchain). It is possible to automate the execution of financial instruments and digital assets using these smart contracts.

[5] `https://augur.net/`.

[6] `https://gnosis.io/`.

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market. We may find in Slamka et al. (2013) a review of some AMMs and their characteristics, describing how they vary in parameters, arbitrage conditions, and design.

In this paper, in order to investigate the design of PMs and also their liquidity management, we build an agent-based model (Anderson et al., 1988; Holland & Miller, 1991; Arthur et al., 1997; Bonabeau, 2002; Blume & Durlauf, 2005; Tesfatsion, 2006; Farmer & Foley, 2009; Steinbacher et al., 2021), which is an extension of the Santa-Fe Institute Artificial Stock Market (Palmer et al., 1994; Lebaron, 2002; Johnson, 2002; Ehrentreich, 2006).[7] We focus on two different AMMs, namely the Logarithmic Market Scoring Rule (LMSR) (Hanson, 2007) and the Liquidity Sensitive Logarithmic Market Scoring Rule (LS-LMSR) (Othman et al., 2013). LS-LMSR is an extension of LMSR in which liquidity varies according to the number of outstanding stocks. We call our model "Logarithmic Market Scoring Rule Artificial Stock Market" (LMSR-ASM). To demonstrate the capabilities of the LMSR-ASM, we present the results of simulations regarding the sensitivity of profits in relation to market characteristics, such as the number of participants, as well as choices made by the creator of a PM, such as rounding methods, and parameter values. We answer five research questions: (1) How to find optimal parameters for market creation; (2) How profits differ under different AMMs; (3) How rounding floating-point calculations affects results; (4) How initial price setting can change market outcomes; and (5) How the sum of prices changes in relation to outstanding quantities in LS-LMSR. In addition, we replicate expected behavior and produce novel results. Our fiddings reveals that, under certain circumstances, the LS-LMSR can be less profitable than the LMSR. We also find that the LMSR-ASM can aid in finding optimal parameters for profit-maximizing a PM. The model also shows that different rounding choices can increase profits without hindering market activity. The LMSR-ASM provides a variety of settings to enable further research.

Our paper connects to the agent-based models of Slamka et al. (2013), Klingert & Meyer (2018) and Brahma et al. (2012). However, our approach differs from theirs due to being the only one evaluating LS-LMSR, as well as adding probability shifts in each period of simulation and offering multiple options for agent behaviors. In particular, our model, the so-called LMSR-ASM, is capable of simulating a market with two stocks, representing a binary PM, for different AMMs, types of agents, and probability trajectories of the underlying event.

It is worth mentioning that our paper also relates to several relevant extensions

[7] The source code for the model is available at `https://github.com/Athosvcc/LMSR-ASM`. The Appendix provide the user guide and offers class descriptions.

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and applications of the Santa-Fe Institute Artificial Stock Market. For instance, Joshi et al. (1998) explores the use of technical trade rules in stock markets. Tay & Linn (2001), and Linn & Tay (2007) update the expectations formation model with fuzzy rules. Zhang & Zhang (2007) investigates the survival of irrational investors in artificial stock markets. Yang et al. (2015), with a new set of parameters, compares the U.S and Chinese financial markets. Yang & Chen (2018) explores the model with different types of agents. Tsao & Huang (2018) explores the issue of survivability and market efficiency, and Huang & Tsao (2018) investigates the effect of evolutionary frequency on forecasting accuracy.

Our work proceeds as follows: In Section 2, we highlight the main ideas behind the PM liquidity management and the design of AMMs. In particular, in Subsections 2.1 and 2.2, we detail the characteristics of the LMSR and LS-LMSR AMMs. In Section 3, we review the Santa Fe Institute Artificial Stock Market as presented in Ehrentreich (2008), which we use as a basis for our work, and we introduce the LMSR-ASM. Section 4 presents the results. We divide this section into five subsections. In Subsection 4.1 we analyze the choices of some parameters that are essential to define the LMSR and the LS-LMSR. In Subsection 4.2, we compare the profits of LMSR and LS-LMSR AMMs in their basic setups. In Subsection 4.3, we evaluate the effect of rounding on profits. In Subsection 4.4, we calculate how much initial prices affect profits. In Subsection 4.5, we check how the sum of prices varies according to outstanding quantities in LS-LMSR. Finally, Section 5 presents the main conclusions of the work.

2. AMMs

As already mentioned, PMs encounter difficulties in knowledge aggregation[8] due to low liquidity, which leads to the necessity of an automated market maker willing to buy or sell stocks at any time for a determined price.

The starting point for the design of AMMs is the concept of scoring rules. Scoring rules are functions that assess the quality of probabilistic forecasts by giving a numerical value that depends on the predictive distribution and the real outcome. A score rule is said to be proper if an agent maximizes their expected score by revealing their truthful probability expectation (Gneiting & Raftery, 2007). A market scoring rule is a sequential implementation of a proper scoring rule. We may describe the recipe for implementing a market scoring rule in the following way. The market

[8]Knowledge aggregation refers to the use of a price system as a mechanism for communicating dispersed information, as shown in Hayek (1945).

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opens with an initial probability distribution of the outcomes. Every trader in the market may change the current probability estimate of the outcome to a new estimate by accepting (and paying) the score rule payment associated with the current probability estimate and receiving the score rule payment associated with the new probability estimation. The market maker only pays the last trader and receives payment from the first trader. However, although this works in theory, it does not seem to be practical to ask for the traders reporting probabilities (Chen & Pennock, 2010). Chen & Pennock (2007) provides a solution to this problem, showing that a market scoring rule is equivalent to a cost function based market maker.

Suppose that a market possesses n mutually exclusive and exhaustive outcomes and the probabilities associated with these outcomes sum 1. Let q_i be the total quantity of security i that pays 1 dollar if the outcome i happens, and let $q = [q_i]$ be the vector of outcomes. An AMM may use a cost function $C(q)$ that records the total amount of money traders have spent as a function of the total number of shares of each security. Thus, a trader who wants to purchase δ shares of security i must pay $C(q_1, \cdots, q_i + \delta, \cdots, q_n) - C(q_1, \cdots, q_n)$ dollars. In general, a trader that wants to buy a portfolio of shares changing the number of shares of the market from q_{old} to q_{new} must pay $C(q_{\text{new}}) - C(q_{\text{old}})$ dollars. As we described before, a market maker using a cost function only pays the last trader and receives payment from the first trader (Pennock & Sami, 2007). Let p be a vector of prices. At any time, the price of security i is $p_i(q) = \partial C(q)/\partial(q_i)$, the cost per share for purchasing an infinitesimal quantity. We may define the price rule $p : \mathbb{R}^n \to [0, 1]^n$ as a differentiable function that maps a vector of quantities to a vector of prices.

There are some desirable properties for cost functions:

1. *Monotonicity* (Hanson, 2003; Othman et al., 2013): For all q and $\overline{q}$ such that $q_i \leq \overline{q}_i$, we have $C(q) \leq C(\overline{q})$. Monotonicity avoids arbitrages in which a trader may create a zero-cost portfolio with no losses and positive gains.
2. *Convexity* (Agrawal et al., 2011)): For all q and $\overline{q}$ and $\lambda \in [0, 1]$, $C(\lambda q + (1 - \lambda)\overline{q}) \leq \lambda C(q) + (1 - \lambda)C(\overline{q})$: Convexity suggests that the market maker should diversify the portfolio, since the cost of the combination of two portfolios q and $\overline{q}$ is never greater than the combination of the costs of the individual portfolios $C(q)$ and $C(\overline{q})$.
3. *Positive homogeneity* (Othman et al., 2013): For all q and $\gamma > 0$, $C(\gamma q) = \gamma C(q)$. Positive homogeneity ensures that if q increases by a factor γ, then the cost also increases by the same factor.
4. *Bounded loss* (Hanson, 2003): For all q with $||q|| < \infty$, $\sup_q[\max_{q_i} - C(q)] < \infty$. A market maker using a cost function with bounded loss can only lose a finite amount to traders.

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5. *Translation invariance* (Agrawal et al., 2011): For all q and scalar α, $C(q + \alpha 1) = C(q) + \alpha$. Translation invariance means that by increasing the quantity q by the same amount in every state of the world, the cost increases by this amount.
6. *Path independence* (Hanson, 2003; Othman et al., 2013): the cost of changing the quantity vector from q to $\overline{q}$ depends only on q and $\overline{q}$, not on the path between them. *Path independence* guarantees the market maker cannot become a money pump: any sequence of trades results in the same overall cost, so agents are not able to generate profits only by placing a set of trades.
7. *Liquidity sensitivity* (Othman et al., 2013): *Liquidity sensitivity* replicates the behavior of stock markets regarding outstanding quantities. The more available stocks a market has, the less a fixed value investment moves prices. It is the opposite of *liquidity insensitivity* that we may mathematically define in the following way: let $1 = (1, 1, \cdots, 1)$, for all q, $p_i(q + \alpha 1) = p_i(q)$.

A relevant result is that convexity, monotonicity and translation invariance happens if, and only if, $\sum_{i=1}^{n} p_i(q) = 1$. It means that in this case, we can interpret prices as probabilities. On the other hand, Othman et al. (2013) shows that no cost function jointly satisfies path independence, translation invariance, and liquidity sensitivity. We must abandon one of the other two properties to create a liquidity-sensitive market maker.

We focus on the following AMMs: Logarithmic Market Scoring Rule (LMSR) and Liquidity-Sensitive Logarithmic Market Scoring Rule (LS-LMSR). The LMSR satisfies the path independence and translation invariance properties. The LS-LSMR satisfies the path independence and liquidity sensitivity properties. Both AMMs satisfy the other properties, namely monotonicity, convexity, positive homogeneity, and bounded loss. We review these market makers in the following subsections. It is worth mentioning that there is a variety of other significant market makers, for instance, the Dynamic Pari-Mutuel Market (Pennock, 2004), and Constant Product AMMs (Wang, 2020), both of which are outside of the scope of this paper.

2.1. The Logarithmic Market Scoring Rule (LMSR)

The LMSR is a Market Scoring Rule formulated in Hanson (2007), widely used in PMs, due to the property of all prices summing to unity.

The general formula for the LMSR is a cost function of the total of assets in the PM, given by:

$$C(\boldsymbol{q}) = b \log \left(\sum_{j=1}^{n} e^{q_j/b} \right), \tag{1}$$

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where $\boldsymbol{q}$ is a vector of quantities, b is a strictly positive parameter that controls liquidity in the market, and n is the number of assets (outcomes) in the market. This cost function represents the total amount of money spent on buying or selling all shares available in the market (Slamka et al., 2013).

The derivative of the cost function yields the price function for infinitesimal quantities, given by:

$$p_i(\boldsymbol{q}) = \frac{e^{q_i/b}}{\sum_{j=1}^{n} e^{q_i/q}} \tag{2}$$

The creator of a PM sets market liquidity via the variable b. The higher b is, the less a fixed value investment shifts prices. Market maker's loss is a function of this parameter, with a worst-case loss of $b \log n$ (Hanson, 2007). The LMSR is modular. Also, it is able to handle stocks with an arbitrary number of outcomes, from binary markets to combinatorial ones. Its only requirement is that assets represent all possible final states. The LMSR achieves maximum loss when the market for an event converges to a corner value before the market closure. In less extreme cases, market maker loss is given by the difference between the entropies of the initial and final distributions.

Due to consistent losses for the market maker, it is not ideal to use the LMSR in for-profit exchanges. It is possible to avoid losses by imposing costs on transactions, for instance, charging for buy and sell orders or cashing in profits. These costs distort price interpretations as forecasts of the probability of each outcome since agents factor these fees into their behavior.

2.2. A Liquidity Sensitive LMSR (LS-LMSR)

The LS-LMSR turns the parameter b of the original LMSR into a function of the outstanding quantities (Othman et al., 2013):

$$C(\boldsymbol{q}) = b(\boldsymbol{q}) \log \left(\sum_{j=0}^{n} e^{q_j/b(\boldsymbol{q})} \right), \tag{3}$$

where

$$b(\boldsymbol{q}) = \alpha \sum_{i} q_i \tag{4}$$

with α being a strictly positive parameter set by the owner of the PM.

When we relax the translation invariance, prices in LS-LMSR do not sum to unity since they do not satisfy translation invariance. Instead, at any quantity vector, the sum of all prices follows strict lower and upper bounds:

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$$1 \leq \sum_i p_i(\boldsymbol{q}) \leq 1 + \alpha n \log n \tag{5}$$

The sum of all prices approaches the lower bound when $q_i \to \infty$ and $q_j = 0$ for $i \neq j$. The upper bound is reached when all quantities are equal, $\boldsymbol{q} = k_1$ for any k_1. When the sum of prices reaches its upper bound, any probability between $\frac{1}{n} - \alpha(n-1)\log n$ and $\frac{1}{n} + \alpha(n-1)\log n$ would be consistent with current prices.

The choice for the parameter α can follow the intuition from traditional betting markets. Since the sum of prices is dependent on α, the creator of the market can set an upper bound for its possible maximum commission, also called vigorish, or the *vig*. For any desired *vig* v, we can guarantee that the maximum commission does not exceed v by setting the parameter α as follows:

$$\alpha = \frac{v}{n \log n}. \tag{6}$$

As an example, for a maximum *vig* of 20%, common in bookmaker markets (Smith et al., 2006), and a PM with two possible outcomes, the parameter α should be set to

$$\alpha = \frac{0.2}{2 \log 2} \approx 0.1442.$$

As we mentioned before, like the original LMSR, the LS-LMSR exhibits bounded loss, but instead of $b \log n$, it is given by the cost function at initial quantities $C(\boldsymbol{q^0})$. It means the market maker can lower their losses by setting initial quantities as low as it is desired. It has a trade-off in initial market liquidity. In a market with $\alpha = 0.05$ and with only one outstanding quantity, an initial bet would cost over \$0.95 (Othman et al., 2013), so even though it is possible to make the worst-case loss approach zero, it may discourage trading.

When it comes to the revenue for markets with the same worst-case loss, the LS-LMSR generates equal or more revenue than the LMSR for the same set of trades, no matter the outcome or outstanding quantities. Worst-case revenue can be presented in closed form and is given by the equation

$$\mathcal{R}(\boldsymbol{q}) \equiv C(\boldsymbol{q}) - \max_i q_i - C(\boldsymbol{q^0}). \tag{7}$$

Thus, the appeal of the LS-LMSR also comes from achieving profits more frequently in relation to the original LMSR, in exchange for a marginal loss in the interpretation of prices as probabilities, being a better choice when running for-profit Prediction Markets.

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3. The Model

This section aims at presenting the model we use to simulate the PMs. Since our model uses the Santa-Fe Institute Artificial Stock Market (SFI-ASM) (Palmer et al., 1994; Lebaron, 2002; Johnson, 2002) as a basis for the LMSR-ASM, in Subsection 3.1, we revisit this model and we review its main ingredients. In Subsection 3.2, we present our model. For the sake of organization, we divide that subsection into three other subsections. In Subsection 3.2.1, we present the stocks available in the LMSR. Subsection 3.2.2 introduces the types of agents that may trade stocks in this market. Finally, Subsection 3.2.3 details the rounding procedure used in the LMSR-ASM.

3.1. SFI-ASM

We adapt the SFI-ASM originally presented by Ehrentreich (2008) to run our PM simulations. This version considers an arbitrary number of traders selected at the beginning of the simulation. Each trader holds one unit of risky stock and 20,000 units of cash. During each period, traders make investment decisions by choosing how much to invest in stocks and how much to keep in cash, which yields a risk-free rate of return r_f.

In the SFI-ASM, agents are myopic, meaning they only consider prices for the next period. They are also homogeneous with respect to their utility function, determining their optimal stock holdings at each period by maximizing a Constant Absolute Risk Aversion (CARA) utility in the form:

$$U(W_{i,t+1}) = -e^{-\lambda W_{i,t+1}}, \tag{8}$$

where λ is a strictly positive parameter, representing the degree of risk aversion, and $W_{i,t+1}$ is the expected wealth of agent i for the next period. Agents face the following budget constraint:

$$W_{i,t+1} = x_{i,t}(p_{t+1} + d_{t+1}) + (1 + r_f)(W_{i,t} - p_t x_{i,t}), \tag{9}$$

where p_t is the price at time t, $x_{i,t}$ is the amount of stock held by agent i at period t, and d_{t+1} is a stochastic dividend the stock pays, given by a mean-reverting autoregressive Ornstein-Uhlenbeck process, of the form

$$d_{t+1} = \overline{d} + \rho(d_t - \overline{d}) + \epsilon_{t+1}, \tag{10}$$

where $\overline{d}$ is the dividend mean, ρ is the speed of mean reversion and ϵ are stochastic shocks following $\mathcal{N}(0, \sigma^2{}_\epsilon)$.

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The solution to the utility maximization problem returns the optimal risky stock holding at any period, which is given by the equation

$$\widehat{x_{i,t}} = \frac{E_{i,t}[p_{t+1} + d_{t+1}] - p_t(1 + r_f)}{\lambda \sigma^2_{t,p+d}}, \tag{11}$$

with $\sigma^2_{t,p+d}$ being the empirically observed variance of the stock's price plus dividend time series, $E_{i,t}[p_{t+1} + d_{t+1}]$ being agent i's expectation of next period's stock price and dividend.

Traders derive their expectations from a set of trading rules that represents unique conditions to each agent, based on fundamental or technical analysis of the dividend process. An example of fundamental rule is "dividend–price ratio is under 1/2". Thus, if the current dividend–price ratio is smaller than 1/2 the agent buys the stock. Otherwise, the agent sells it. A technical rule is, for instance, "the 25-period moving average of the stock price is greater than the current price". If the 25-period moving average is smaller than the current price, the agent buys it. Otherwise, the agent sells it. Note that these are Boolean conditions. According to them, and considering the fitness values for each rule, agents make their forecasts. Different expectations represent how heterogeneous agents process the same information set available to all traders. In the SFI-ASM, agents can learn by changing their trading rules by feedback learning, observing which rules produce better forecasts over time, and a genetic algorithm, which enables agents to change rules via mutation and cross-over processes.

3.2. The LMSR-ASM

As already mentioned, the LMSR-ASM adapts the model presented by Ehrentreich (2008) to PM simulations. Simulations start with a previously selected number of agents carrying a predetermined quantity of cash. At each period, agents update their optimal demand for stocks by maximizing a CARA utility. Agents buy or sell stocks according to the difference between their optimal demand, resulting from the maximization problem, and the number of stocks they currently have.

Event probability is either arbitrarily chosen at the beginning of a simulation or defined by one of the generative processes available. We discuss this in detail in Section 3.2.1. Each agent has a forecast of the probability of the underlying event. This forecast relies on the event probability given by the simulation, plus a random element unique to each agent, following a normal distribution $\mathcal{N}(0, 0.025)$ with zero mean and 0.025 variance, as defined in Slamka et al. (2013). This variation aims at modeling the fact that agents build beliefs according to noisy signals with a Gaussian distribution, particular to each one (Das, 2005).

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A simulation run has the following workflow. We consider that a market follows either the LMSR AMM or the LS-LMSR AMM. In the first period, we arbitrarily set initial quantities to create liquidity, when in LS-LMSR, or to adjust initial prices, in both cases. These initial quantities are *artificial* stocks. The market maker holds these stocks, and since they only receive its payment at the end of the simulation, it does not affect profits. The LS-LMSR presents liquidity problems without initial quantities, since, according to Eqs. (3) and (4), the cost function is not well defined at $\boldsymbol{q} = (0, 0)$ and one-unit bet prices being costly at the point of discouraging trades (Othman et al., 2013), making these *artificial* stocks necessary to kickstart the market. Regarding price adjustment, we create stocks until the first-unit bets reach the chosen value. Stocks creation is a practical way to affect prices without changing the behavior of cost functions.

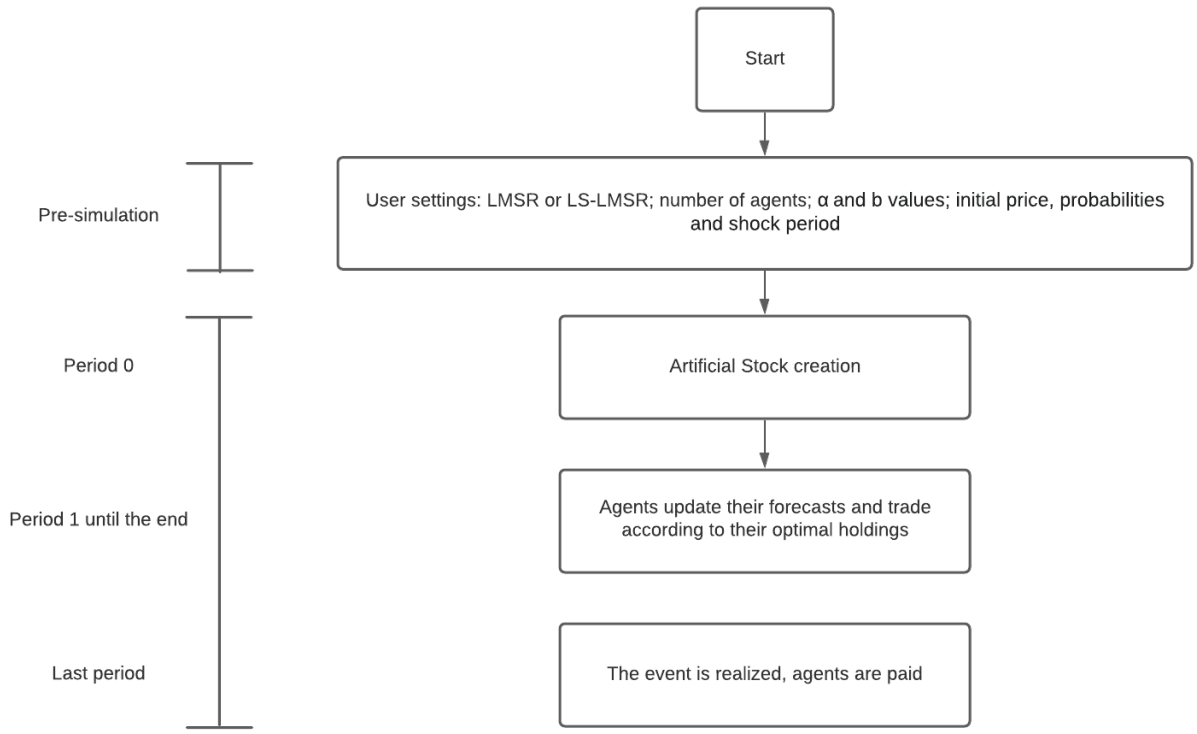

Figure 1. Flowchart of the LMSR-ASM.

After this first period, agents can trade stocks according to their forecasts. In the last period, the AMM closes the market paying the agents with winning stocks according to their holdings. Figure 1 summarizes how the model works.

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3.2.1. Stock Characteristics and Behavior

There are two types of stocks. The *Yes* stock pays \$1 if the underlying event happens. The *No* stock pays \$1 if the underlying event does not occur. Both stocks are related to the same event, with a probability distribution set at the beginning of the simulation. The market maker sets an initial price for the stocks, achieved by the creation of *artificial* stocks, as explained before. Stocks also have a liquidity parameter, b for the case of LMSR or α for the LS-LMSR.

As we mention in Section 2, stock prices in LMSR sum to 1, while in LS-LMSR, the sum of the two stocks varies according to outstanding quantities, with an upper limit of $1 + \alpha n \log n$. Since we consider events with only two possible outcomes, this upper limit is $1 + \alpha \times 2 \times \log 2$ for the case of the LS-LMSR.

Furthermore, it is also possible to include a probability shock at an arbitrary period, in which the probability of the event changes to a preset value. The shock represents a situation in which probabilities change drastically, for example, the likelihood of a football team winning a two-game series after losing the first game.

Stock probability distributions may follow a *fixed* or a *random walk* specifications. In the case of a *fixed* specification, the underlying probability does not change unless a shock is specified. On the other hand, in the case of a *random walk*, at every step, the event's probability changes in relation to arbitrary shocks that follows a normal distribution $\mathcal{N}(0, 0.025)$.

Figure 2 illustrates how each probability setting behaves in a 100 period run. In the fixed set up, the YES event starts with probability 0.7 and drops to 0.3 after a shock at period 50. The Fixed setting is convenient to model the behavior of a stable event, e.g., the probability of a candidate election, considering a week with no significant news regarding the race. On the other hand, the Random walk setting may simulate richer processes.

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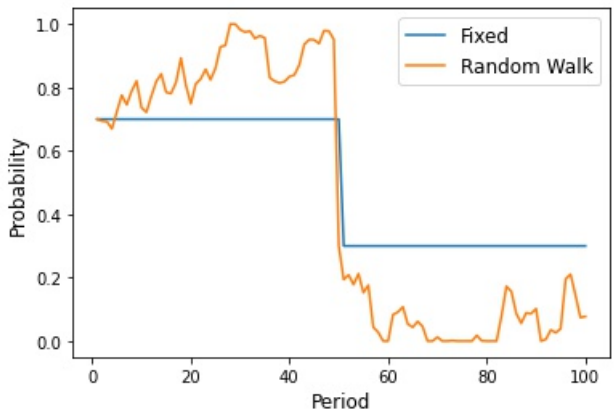

Figure 2. A demonstration of how different probability processes vary over time.

In the final period, the market closes according to the following rule: if *probability* > 0.5, *Yes* stocks payout; otherwise, *No* stocks payout. Therefore, we may summarize the market behavior as follows:

1. At the start of the simulation, quantities are created to set initial prices and liquidity;
2. Exchanges follow either LMSR or LS-LMSR, as predetermined, with parameters respectively given by b or α;
3. Probability distribution is updated at every step according to the process chosen (fixed or random);
4. At a given period, the underlying event's probability changes from its previous value to another predetermined one;
5. At the end, the underlying event happens, and agents receive their payments depending on whether they invest in the correct stock.

3.2.2. Agent Characteristics and Behavior

In this model, we accommodate two types of agents: *ideal* and *noisy*.

The *ideal* agent is a perfect foresight agent, used as a baseline. This agent knows the real probability of an underlying event. This agent's expected payoff for one stock, according to its own forecast, is $E(f_i) = 1 * (f_i - p_s) + 0 * (1 - f_i) = f_i - p_s$, where f_i is agent i's forecast and p_s is the price of the stock the agent is buying. Therefore, the agent has a positive expected payoff for any price $p_s < f_i$. For ideal agents, f_i is the true probability of the event, and they buy stocks until the price

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matches the probability.[9]

The *noisy* agent creates forecasts according to noisy signals they receive regarding the event, given by the actual probability adjusted by a random variable unique to each agent, which follows a normal distribution $\mathcal{N}(0, 0.0025)$. They are utility-maximizing, buying, and selling stocks only until their current stock holding equals their optimal holdings that depend explicitly on their noisy predictions. Following the SFI-ASM specification in Section 3.1, these agents are myopic, meaning they only consider prices for the next period and homogeneous with respect to their utility function, determining their optimal stock holdings at each period by maximizing a CARA utility, shown in Eq. (8). Since our analysis is on PM trading and AMMs, not portfolio decisions, we set the value of the interest rate to zero. Furthermore, since the PM stocks do not pay dividends, the agents' budget constraint differs from Eq. (9). In this case, we get

$$W_{i,t+1} = W_{i,t} + x_{i,t}(p_{t+1} - p_t), \tag{12}$$

where p_t is the price at time t and $x_{i,t}$ is the amount of stock held by the agent i at period t. By solving the utility-maximization problem, we find the optimal risky stock holding at any period as in

$$\widehat{x_{i,t}} = \frac{E_{i,t}[p_{t+1}] - p_t}{\lambda \sigma^2_{t,p}}, \tag{13}$$

with $\sigma^2_{t,p}$ being the variance of the stock, which follows a Bernoulli distribution, and $E_{i,t}[p_{t+1}]$ being agent i's forecast.[10] Since optimal stock holdings are dependent only on the difference between their forecast and the current price, both of which have an upper bound of 1, due to being related to probabilities, the agent's demands rarely exceed 2. In the edge case, with price 0 and an agents forecast of 1, the optimal stock holdings for an agent with $\lambda = 0.3$ would be $\frac{1-0}{0.3\times 1}$. It does not seem to affect convergence, even though it can make prices volatile in a simulation with a small number of agents due to few outstanding stocks.

If the agent's forecast of the probability of the underlying event is higher than the current stock price and it does not hold *No* stocks, it buys *Yes* stocks according to its optimal demand. If the agent's forecast of the probability of the underlying

[9]Note that this is true for both LMSR and LS-LMSR. The fact that prices in LS-LMSR do not sum to unity does not affect agents' portfolio decisions, only the price of the stock they buy does.

[10]Since prices are equivalent to probabilities, the term represents the agent's probability forecast and price expectations for the next period

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event is lower than the current stock price and it does not hold *Yes* stocks, it buys *No* stocks according to its optimal demand. In case the agent's prediction is not in agreement with its stock holdings, the agent changes its current holds until prices change to match its forecast. Table 1 summarizes the agent's behavior. Even though prices in the LS-LMSR do not sum to unity, since the payoff is still $1, each agent's expected return remains unchanged. Therefore, switching AMMs does not change how agents make choices.

	forecast > price	*forecast < price*
holds Yes stocks	buys Yes according to optimal demand	sells all Yes stocks
holds No stocks	sells all No stocks	buys No according to optimal demand

Table 1. The Agent's Behavior in the LMSR-ASM.

3.2.3. Rounding procedure

At each period, agents asynchronously determine their demands, as shown in Section 3.2.2, and then interact with the market maker. Agents can only buy integer stocks. It frequently leads to the necessity of rounding prices. Rounding is also necessary due to floating-point imprecision. The floating cost would allow using the market maker as a money pump. While using floating-point values is also feasible in a simulation setting, it is not recommended in real settings.

In simulation settings, rounding works in two ways: either the cost of an order is closer to the nearest two or, as used in Berg & Proebsting (2009), six decimal digits. The market described by Berg & Proebsting (2009) has a distinct transaction interface, with agents investing determined values and, in return, receiving a rounded number of stocks. In our model, agents order the number of demanded stocks and pay accordingly. Hence, agents pay fracionts of a cent when the AMM rounds prices to the nearest six decimal digits. In a real setting, while users receive the reports of their balances rounded down to the nearest cent, the AMM keeps (internally) the actual values of 6 decimal digits in order to implement transactions.

The AMM handles differently selling and buying stocks. The AMM rounds down prices when agents are selling and rounds up when they are buying. We can see this as a form of "breakage" or a bid-ask spread. The rounding behavior helps to offset part of the market maker's loss. With breakage, there is a change in the path independence characteristic of LMSR. Ideally, an agent could buy and immediately resell their shares without incurring losses. In the model, however, this action possibly leads to small losses.

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4. Results

In this section, we explore the performance of the LSMR-ASM through a series of simulations. Table 2 shows the reference values of the parameters used in our simulations. In Subsection 4.1 we support the choice of the parameters $b = 76.13$ and $\alpha = 0.04$ presented in this table by varying both parameters and comparing the variations in the trading volume and the market profits using both the LMSR and LS-LMSR algorithms for the AMM. In Subsection 4.2, we estimate the profit differences between LMSR and LS-LMSR AMMs. In Subsection 4.3, we evaluate the effect of rounding over the profits. In Subsection 4.4, we calculate how much initial prices affect profits. In Subsection 4.5, we check how the sum of prices varies according to (outstanding) quantities in LS-LMSR. We may modify some of the simulation parameters to deal with the problem explored in each subsection. Unless otherwise stated, we use the reference values of Table 2 in all simulations.

Except for Subsection 4.1, where we intend to find the parameters b and α that we use elsewhere, we run the simulations using both ideal agents and noisy agents. In the first case, the simulations are deterministic, so we perform only one run. In the second case, which involves stochastic factors, we perform 100 runs for each simulation exercise. The use of 100 runs presents a good trade-off between the need to produce credible results and computational costs.

Table 2. Reference values for parameters in LMSR-ASM simulations.

Parameter	Value
Number of periods	100
Probability process	Fixed
b (LMSR)	76.13
α (LS-LMSR)	0.04
Initial price	0.5
Initial quantities	(50, 50)
Initial probability	0.8
Period of probability shock	50
Probability after shock	0.2
Risk aversion	0.3

The fixed probability process shows how AMMs and agents behave in an ideal market, so it is possible to analyze market characteristics independently of random shocks. The same rationale applies to the interest rate.

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The planned shock at period 50 enables us to understand the market's reaction to a sudden change in probabilities after the values have already stabilized with respect to previous knowledge. Probability values are distant from each other, and the initial price highlights the movement of price convergence in relation to probability. Risk aversion is as set in Ehrentreich (2008), the Santa-Fe Institute Artificial Stock Market, used as the basis for the LMSR-ASM.

4.1. Optimal values for parameters b *and* α

To analyze optimal values for b and α, we run simulations varying both parameters, as well as the number of participants in a market. The parameter b varies from 15 to 150, in increments of 15, and the parameter α varies from 0.01 to 0.10, in increments of 0.01. We then compare the variations in the trading volume and the market profits during simulations in order to understand how LMSR and LS-LMSR models behave when these two parameters change. We consider here only simulations with noisy agents since we intend to determine the parameters we use in the rest of this paper. Besides, we believe these values, once determined to the noisy case, are more suitable for reflecting on real-world situations.

We present the results of the simulations of the LMSR-ASM model using LMSR AMM with noisy agents in Figure 3 for many different choices of the parameter b. Panel (a) shows the effects on trading volume, and panel (b) shows the effects on market profits. In this case, it is possible to analyze how trading volume and market profits change with a varying number of agents and with different choices of the parameter b. Trading volume increases both with b and with the number of agents, while market maker profits decrease with respect to b and increase with respect to the number of agents. There is no optimal rule for choosing the parameter b. Also, it is possible to increase profits by decreasing this parameter, but it might also excessively reduce liquidity that prices do not precisely reflect the underlying probability.

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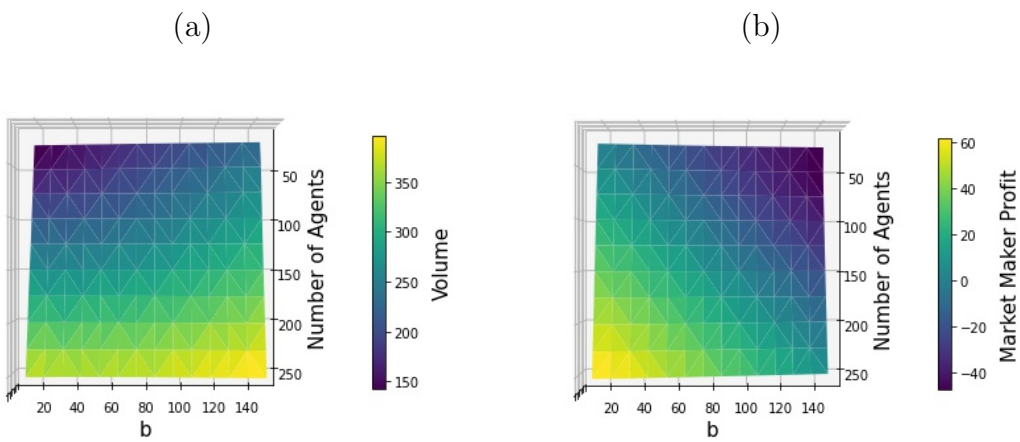

Figure 3. Parameter analysis in the LMSR model with noisy agents. (a) Effect on trading volume; (b) Effect on market profits.

Figure 4 shows a case where prices do not converge to the probability due to low liquidity. Since agents update forecasts every period following a normal distribution with zero mean, simulating for periods longer than 100 would not change the lack of convergence. Prices would continue to vary around the true probability value, as shown in Figure 4. Since buying a few stocks leads to higher prices, we see wider price gaps. It is related to the changing perception of the agents. The prices vary around the true probability, but they do not converge and are highly volatile. In any case, batch simulations can help market creators set their initial parameters according to the desired volumes and profits.

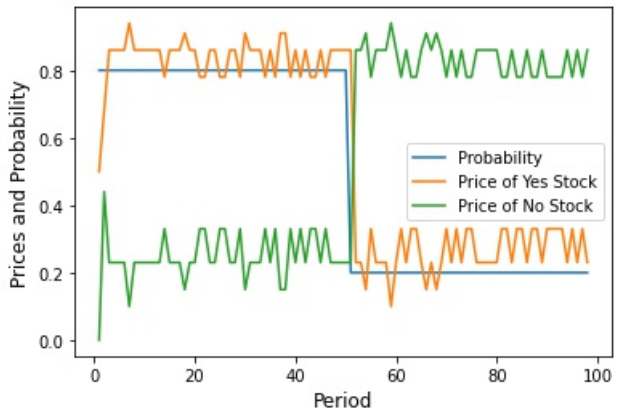

Figure 4. An example of lack of convergence to probability when liquidity is low ($b = 2$).

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Knowledge aggregation is the most relevant factor in choosing model settings for prediction markets that do not have profit maximization as an end goal. In this scenario, trading volume is the key variable responsible for small differences in agents' perception of the probability of the underlying event. When liquidity is high, it is difficult for participants to affect prices. Despite, play money markets can compensate for this situation without drawbacks by distributing more tokens to participants, thus being more efficient in terms of knowledge aggregation. On the other hand, for-profit markets deal with the trade-off between volume, related to the liquidity parameter b, and market profits. Especially when considering the LMSR, which is rarely profitable without additional fees, lower liquidity can completely undermine trade as agents factor extra costs into their expectations, distorting market prices and decreasing overall participation.

Figure 5 shows the outcomes of the simulations of the LMSR-ASM model using LS-LMSR AMM with noisy agents. In this case, as liquidity and profits are (directly) linked by the parameter α, there is an optimal choice for each market, a value that maximizes the market maker's profits. Lower α values decrease worst-case loss, but also decrease liquidity to the point of reducing volume and market values, which, in turn, affects profits. However, higher α values also decrease volume. Agents trade less because prices are not directly translated to probabilities. As discussed in Section 2.2, when the sum of prices reaches its upper bound, given by $1 + \alpha n \log n$, any probability between $\frac{1}{n} - \alpha(n-1)\log n$ and $\frac{1}{n} + \alpha(n-1)\log n$ is consistent with stock prices.

(a) (b)

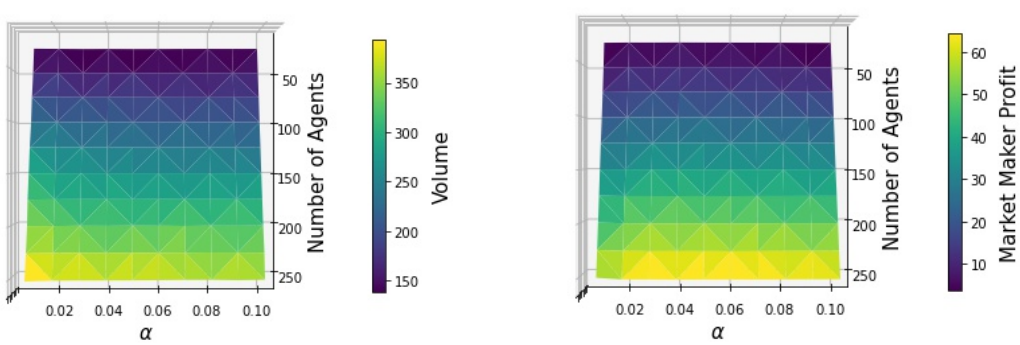

Figure 5. Parameter analysis in the LS-LMSR model with noisy agents. (a) Effect in trading volume; (b) Effect in market profit.

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Trading volume increases in relation to the number of agents. Also, it varies slightly with respect to the values of α, mainly due to different seeds in the behavior of the noisy agents. In panel b of Figure 5 we can see that market maker profits increase with α until it reaches its maximum value, at $\alpha = 0.04$. Thus, this value generates the maximal profit and with the *vig* of $0.04 \times 2 \times \log 2 \approx 0.0554$, or 5.54% of the market's revenue. It is worth mentioning that this value of α may not be consistent with different types of agents or probability behaviors. Still, it is relevant for further simulations and to demonstrate the model's ability to find an optimal value under a specific setting. The possibility of finding an optimal value for the parameter α using simulated data is novel and enables PM creators to maximize profits by considering the expected characteristics of the market.

Since we cannot find an optimal value for b, but only a trade-off between volume and profit, we choose b in order to be equivalent to α with respect to losses. Thus, assuming that $\alpha = 0.04$, we choose $b = 76.13$, since with this value, these parameters (b and α) are equivalent with respect to losses, and they both lead to the same worst-case loss of ≈ 52.77.

4.2. Comparison between LMSR and LS-LMSR profits

To evaluate the differences between the two choices of automated market makers (LMSR and LS-LMSR), we vary the probability after shock as well as the initial probability, from 0.1 to 0.9, inclusive. Figure 6 shows how the AMMs react under this large range of possibilities. Each step of the figure represents a respective specification of the probability after the shock, from 0.1 to 0.9. The simulations within each step correspond to changes in the initial probability. As before, it ranges from 0.1 to 0.9. We use ideal agents for baseline and noisy agents to simulate a market with uncertainty.

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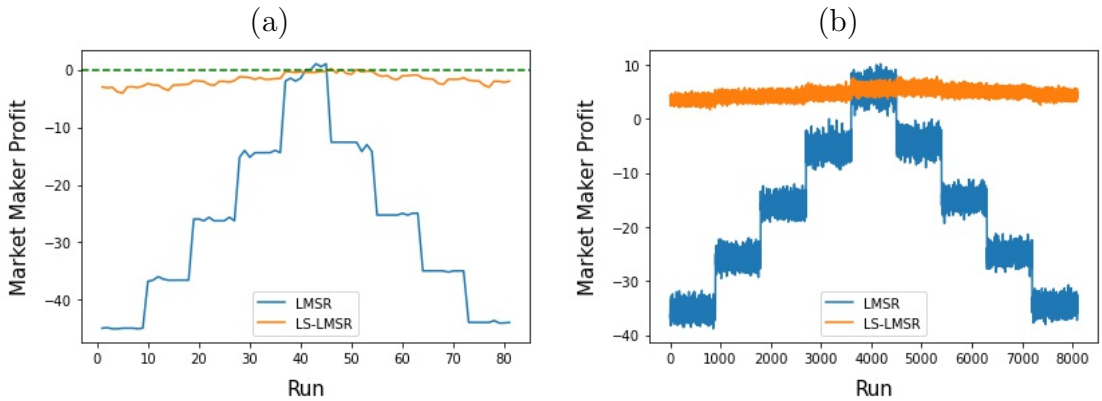

Figure 6. Comparison between LS-LMSR and LMSR AMMs using probabilities from 0.1 to 0.9. After shock probabilities change between steps and initial probabilities change within steps. The dotted green line separates profits and losses. (a) Ideal agents; (b) Noisy agents.

Figure 6 also shows that the LS-LMSR consistently presents higher profits than the LMSR. However, in specific cases, LMSR and LS-LMSR profits converge. It happens when after shock probabilities are close to 0.5, which is similar to initial prices (the middle of the graph). Due to the fact that the set of trades are not equal among simulations, in some runs, the LMSR is more profitable than the LS-LMSR. Under these so-called edge situations, LMSR can be more profitable than the LS-LMSR. It is a consequence of the behavioral changes caused by the different liquidities. LS-LMSR revenue is always higher for an equal set of trades. However, agent incentives are not the same under different AMMs − and, therefore, trades are not equal. For situations that diverge from edge cases, LS-LMSR presents higher profits, reduces variance, increasing the predictability of a market's result.

Simulations with noisy agents present higher profits than with Ideal agents because the market maker revenue increases with the number of trades, which are higher in noisy agents' simulations due to agents constantly updating their forecasts. As already explained, for simulations involving stochastic factors, each setting is used in 100 runs. For deterministic simulations − as the Ideal agents case − we execute only one run. Therefore, the noisy agents' simulations has 100 times more runs than the one with Ideal agents. Since the same settings are used, the simulations are directly comparable.

4.3. Rounding effects

Due to floating point imprecision, we need to adapt the cost calculations in a way that the market maker does not become a money pump (Berg & Proebsting,

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2009). The way a PM creator decides to make these adaptations can reflect on PM revenues. We evaluate two different rounding methods, rounding costs up by the nearest of two decimal digits or by the closest of six decimal digits, as used in Berg & Proebsting (2009). On the one hand, this difference in rounding does not affect behavior significantly. On the other hand, it has a discernible effect on profits, especially when the amount of trading is large.

Figure 7 shows the results of rounding effects in PM profit. We run the model under the same settings as the previous test, with the probability after shock and underlying probabilities changing from 0.1 to 0.9.

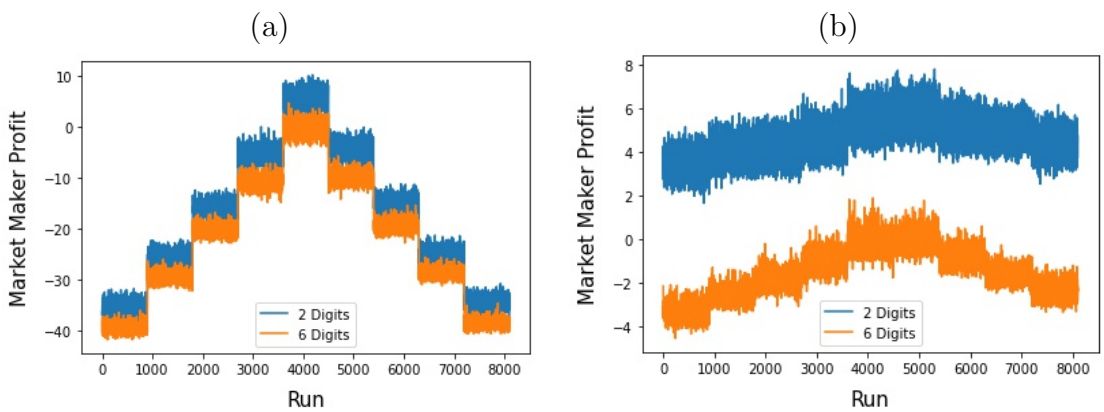

Figure 7. Evaluation of rounding methods: 2-digit rounding vs 6-digit rounding. (a) LMSR; (b) LS-LMSR.

As expected, rounding to 2 decimal digits increases profits in both LMSR and LS-LMSR. The LMSR varies more in general in relation to initial conditions, as shown in Section 4.2. The LS-LMSR presents significantly less variance in profits concerning initial settings. In any case, rounding to 2 decimal digits allows profits to increase in a manner that does not affect market participation significantly.

4.4. How important are initial prices

As presented in Hanson (2007), we can minimize the expected cost of running a market by setting the initial report (price) equal to the market creator's beliefs. To evaluate the impact of price setting, we run the model with a fixed probability. Also, we vary initial prices and the probability after the shock. The batch run ranges from 0.1 to 0.9 for both variables, using Ideal and noisy agents. Figure 8 shows these results.

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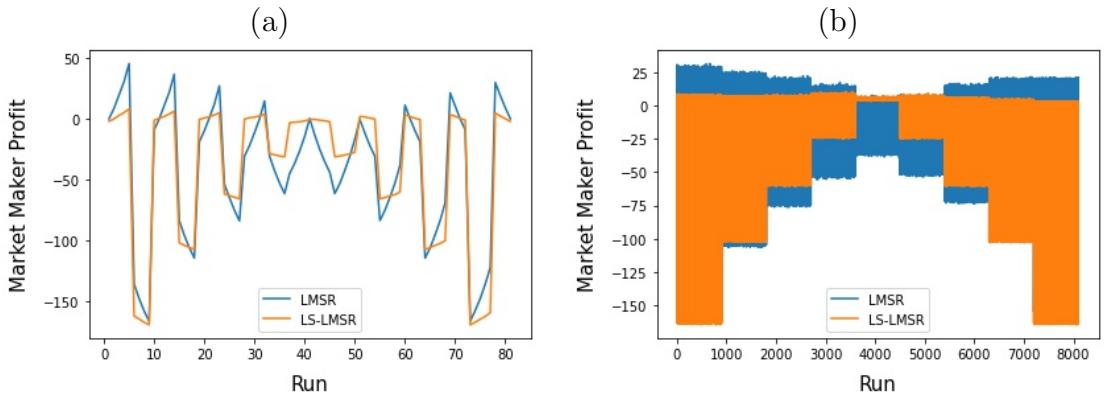

Figure 8. How initial price setting affects profits. (a) Ideal agents; (b) Noisy agents.

The simulation shows that the LS-LMSR only presents higher profits than the LMSR when we set prices to 0.5. These results are in the middle of the graph. Positive spikes occur in runs where the market maker sets initial prices to the correct probability. Negative spikes occur when the opposite happens. Simulations with noisy agents confirm the phenomenon: when initial prices approach corner values, the maximum loss in LS-LMSR is greater than in the LMSR, with the opposite happening when initial prices converge to equality. It can be relevant in PMs in which the market creator has strong priors regarding the final probability. In these cases, using LMSR - and setting initial prices accordingly - might increase profits. We can also observe that setting the initial prices increases profits variance. In the beginning, when prices are not equal, the possibility of the LMSR being more profitable than the LS-LMSR is unexpected and a novel result, showing how the LMSR-ASM can extend our understanding of AMM characteristics.

4.5. Sum of Prices and Quantities in LS-LMSR

The sum of prices in LS-LMSR reaches its maximum when the number of assets is equal and approaches unity as the quantity of one of the stocks goes to infinity. Figure 9 demonstrates how the sum changes as the ratio between stocks changes for three different values of the parameter α, the upper and lower bounds used in previous simulations, and the profit-maximizing value of $\alpha = 0.04$:

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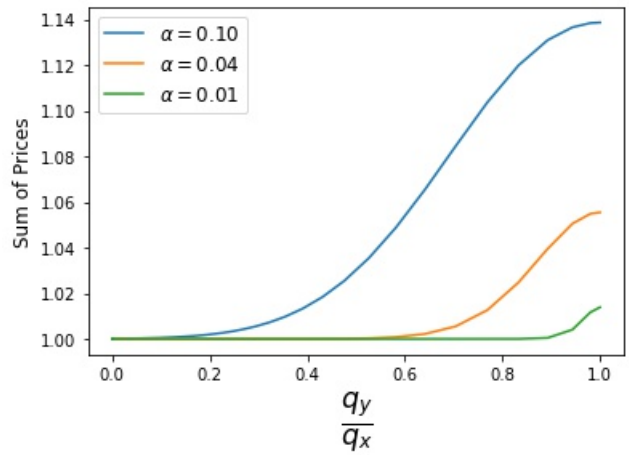

Figure 9. Changes in the sum of prices in LS-LMSR, according to the ratio between stocks, for different values of α.

The plotted values show another characteristic of the LS-LMSR and its sensitivity regarding the choice of the parameter α: not only the parameter defines the maximum profit possible, but it also sets how relative quantities affect profitability. While a market with $\alpha = 0.04$ and a 2:1 ratio has almost converged to the lower bound of profitability, the same quantities still lead to a *vig* of approximately 4% with $\alpha = 0.10$. It leads to a rule of thumb in market creation: when convergence to a corner price is expected, a higher alpha is needed to generate consistent profits. Analysis of the underlying events is relevant in market creation.

5. Conclusion

PMs are efficient in aggregating information regarding the likelihood of uncertain events. Due to liquidity problems, most PMs utilize AMMs to guarantee agents can buy and sell shares at any moment in time. A variety of AMMs exists, with different cost functions and underlying behavior, which, in turn, affect agent participation. The most studied mechanism is the Logarithmic Market Scoring Rule (LMSR). The LMSR can be altered to be liquidity-sensitive (LS-LMSR).

We adapted the Santa-Fe Institute Artificial Stock Market to run PM simulations under different initial settings to address how agent behavior and market structure change profits for different types of AMMs and their respective settings.

The model, called LMSR-ASM, demonstrates it is possible to set an optimal parameter value for profit maximization when using the LS-LMSR. The LMSR parameters, in turn, depend on subjective evaluation of a market's creator. It comes

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from the trade-off between profits and market volume. Other results are that market maker loss is sensitive to initial prices, making it worthwhile for owners of PMs to start markets with prices as close as the true probability in the case underlying events do not converge before realization. In some cases, setting initial prices might increase profit variance and make the LS-LMSR less profitable than the LMSR. We also address different calculation rounding options, showing that we can use a (necessary) solution to floating point imprecision to increase profits without affecting market behavior significantly.

The LMSR-ASM reproduces the expected behavior, showing how profits decrease for higher values of the parameter b in LMSR and how the LS-LMSR is more profitable under certain circumstances. In addition, the model also produces novel results, such as showing the possibility of parameter optimization in LS-LMSR, and how the LS-LMSR reduces variance in profits when initial prices are \$0.5, but can be less profitable for a range of other initial prices.

When using the LS-LMSR, a market creator can set liquidity such that profits are more likely. The market creator must ensure that profits can hinder the PMs capacity of aggregating information since transaction costs change agent behavior, leading to less convergence. Losses are not undesirable in play money markets, so liquidity should be set low enough that agents can influence prices but at the same time not so low that prices change significantly with small orders. It makes it so PM managers should evaluate parameters on a case-by-case basis. The LS-LMSR has the highest capability of generating profits when initial prices are equal to 0.50. In this setting, where there is no additional market fees, the LMSR generally leads to losses for the market maker.

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